TECHNICAL PROJECT REPORT // DECISION-TO-ACTION ENGINE

LIGHTS DOWN

A 90-Second Movie Night Decision-to-Action Engine with Brain Battery Calibration, Least-Misery Group Consensus, and the 20-Minute Rule Backup Reel

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GITHUB REPOSITORY
https://github.com/priyansh4336/lights-down

TECH STACK
Vite 6 + React 18, WebGL2 Shaders, PWA, Vitest

1. Executive Summary & Problem Statement

Modern streaming viewers face a well-documented behavioral bottleneck: households spend an average of **35 to 45 minutes scrolling** across fragmented catalogs, only to experience decision fatigue, pick a high-friction title out of exhaustion, and **bail within the first 20 minutes**. Existing movie recommendation apps treat movie night as an open-ended catalog browsing problem—returning 10 to 50 cards or requiring endless Tinder-style swiping that amplifies choice overload.

Lights Down reframes the problem from *catalog discovery* to a **90-second decision-to-action engine**. Instead of asking users to browse genres, Lights Down calibrates their evening across four concrete operational constraints (Vibe Beam, Brain Battery 1–5, Time Budget in minutes, and Crew Size), computes a genre-diversified **Rule of 3 Shortlist**, locks the commitment via a tactile **Perforated Ticket Tear**, and ships every screening with a pre-selected **Backup Reel ("The 20-Minute Rule")** and a back-solved screening timeline.

> **Core Thesis:** Reducing a household's shortlist to 3 genre-distinct candidates while pre-computing a lower-cognitive-load Plan B eliminates both pre-play analysis paralysis and mid-stream bail regret.

Key Engineering Highlights in v2.0

- **Zero-Config Offline + Live TMDB Architecture:** Ships with 60 curated Hollywood, Bollywood, Regional Indian (Malayalam, Tamil, Telugu, Kannada), and World Cinema films enriched with verified TMDB metadata (`src/data/tmdb-cache.json`) and optional live Indian streaming badges via JustWatch/TMDB.
- **Provable 20-Minute Rule & Telemetry:** Tracks real decision time (seconds from setup start to ticket tear), runs a live 20-minute session check-in timer with optional browser notifications, and computes a data-driven **Regret Report** once 5+ nights are logged.
- **Multi-Modal Group Consensus:** Supports both same-room Pass-the-Phone **Hot Seat** and zero-backend remote **Crew Link** voting (`#crew=...` URL hash + compact `base64url` ballot codes) aggregated via Least-Misery math.

2. Market Landscape & Competitor Comparison

We audited seven representative movie pickers and social discovery tools (Chooza, Swyf, WatchLitz, Project Flow, Movievia, Movik, and Letterboxd). While all seven focus on pre-selection filtering or social cataloging, none address what happens *after* a pick is made or account for cognitive exhaustion at night.

Product	Primary Mechanism	< 90s Target	Brain Battery	Backup Reel (20m Rule)	Back-Solved Timeline	Group Least-Misery	Bail & Regret Learning
Chooza	AI questionnaire	No	No	No	No	No	No
Swyf	Group swipe matching	No (~12m)	No	No	No	Pairwise intersection	No
WatchLitz	Catalog & watchlists	No	No	No	No	No	No
Project Flow	Mood/genre filter grid	No	No	No	No	No	No
Movievia	Similarity recommendations	No	No	No	No	No	No
Movik	Swipe deck recommender	No	No	No	No	No	Likes only
Letterboxd	Social diary & lists	No	No	No	No	No	Manual diary
Lights Down	**Decision-to-Action Booth**	**Yes (<90s)**	**Yes (1–5 scale)**	**Yes (Plan B + Timer)**	**Yes (Split-Flap)**	**Yes (0.6 min + 0.4 mean)**	**Yes (Patience + Genre)**

3. Standard Operating Procedure (SOP): The 6 Differentiators

1. **Backup Reel ("The 20-Minute Rule"):** Every primary pick is paired with a deterministic Plan B that is strictly shorter, has a faster hook (`hookMinutes`), demands equal or lower Brain Battery, and pivots to a different primary genre. At minute 20 of playback, Lights Down prompts: "*20 minutes in. Still with it?*"
2. **Time-Fit & Double Feature Back-Solver:** Given a user's available window (90 to 240 minutes), the engine rounds start time to the next 5-minute mark + 10 minutes settling time, computes the 20-minute bail checkpoint, and automatically offers a lower-battery second film with a 10-minute intermission when the budget allows (≥ 200m).
3. **Brain Battery Calibration (1–5):** Replaces abstract genre checkboxes with cognitive bandwidth—from Level 1 (*Dead Battery: comfort watch*) to Level 5 (*Full Cinema Brain: dense masterworks like Primer or Stalker*).
4. **Group Consensus (Hot Seat + Crew Link + Veto Tokens):** Supports 1–4 viewers via private Pass-the-Phone voting or shareable `#crew=...` URLs, scoring films via Least-Misery so no single member is miserable, while granting each player exactly 1 Veto Token.
5. **Rule of 3 + Ticket Tear Commit Ritual:** Never renders more than 3 cards simultaneously and enforces primary genre diversity across slots. Tearing the perforated ticket stub triggers a 90-second settling countdown.
6. **Regret-Aware Learning Loop:** Logging an outcome as *Loved* (+2), *Finished* (+1), or *Bailed* (-2) updates persistent genre weights and adjusts the user's `patience` index (dropping by 0.12 on early <20m bails to prioritize fast hooks next time).

4. POS (Platform Operating System) Engine & Mathematical Formulation

The recommendation and scheduling pipeline in `src/utils/engine.js` is implemented as a pure, deterministic 9-stage functional pipeline with zero external runtime dependencies.

```
[1. Inputs: Mood, Battery B∈{1..5}, Time T∈[90..240], Crew C] │ ▼ [2. Hard Filter + 3-Step Graceful Relaxation (guarantees ≥ 6 candidates)] │ ▼ [3. Single-Viewer Scoring: Mood(0.35) + Genre(0.25) + Rating(0.15) + Hook(0.15·M_B·M_P) + Fresh(0.10) - BailPen] │ ▼ [4. Group Least-Misery Aggregation: Score_crew = 0.60 · min(S_i) + 0.40 · mean(S_i) (Vetoed removed)] │ ▼ [5. Rule of 3 Diversifier: Top 3 candidates with distinct primary genres (G_1 ≠ G_2 ≠ G_3)] │ ▼ [6. Backup Reel Selector: argmax Score s.t. Runtime < Pick, Hook ≤ Pick, Battery ≤ Pick, Genre ≠ Pick] │ ▼ [7. Timeline Back-Solver] ──► [8. Real 20-Min Timer] ──► [9. Night Log & Regret Report Learning]
```

Stage-by-Stage Mathematical Specification

Stage 1 — Hard Filter & Relaxation Hierarchy: Given candidate pool M excluding vetoed IDs V:

- **Strict Pass:** $m.\text{runtime} \le T \land m.\text{brainBattery} \le B \land C \in m.\text{crewFit} \land \text{mood} \in m.\text{moods}$.
- **Relaxation 1 (if $|S| < 6$):** Drop mood overlap constraint while preserving T, B, and C.
- **Relaxation 2 (if $|S| < 6$):** Relax cognitive cap to $m.\text{brainBattery} \le B + 1$.
- **Relaxation 3 (if $|S| < 6$):** Relax runtime ceiling to $m.\text{runtime} \le T + 30\text{ min}$.

Stage 2 — Individual Film Utility Score $S(m, u)$:

```
S(m, u) = 100 × [ 0.35·I(mood ∈ m.moods) + 0.25·clamp(0.5 + 0.08·Σ w_g, 0, 1) + 0.15·(m.rating / 10) + (0.15·λ_battery·λ_patience)·((30 - m.hookMinutes)/28) + 0.10·Freshness(m) - BailPenalty(m) ]
```

where $\lambda_{\text{battery}} = 1.35$ when $B \le 2$ (else 1.0), and $\lambda_{\text{patience}} = 1.35$ when $\text{patience} < 0.50$ (else 1.0). Previously bailed movies incur a direct penalty of 0.35 and shared genres incur 0.06 per historical bail.

Stage 3 — Group Least-Misery Consensus: For a crew of K players with Hot Seat or Crew Link ballots:

```
S_group(m) = 0.60 × min_{i ∈ 1..K} S_i(m) + 0.40 × ( (1/K) · Σ_{i=1..K} S_i(m) )
```

Each thumbs-up adds $+18$ to $S_i(m)$ and each thumbs-down subtracts -22. Any film receiving a Veto Token is immediately pruned from the candidate set for all players.

5. System Architecture & Modular Codebase Structure

In Phase 1, the monolithic v1 application was refactored into a clean, lazy-loaded modular architecture separating state hooks, screen views, style layers, and WebGL/audio effects. Production Rollup chunk splitting keeps every bundle well below the 500 kB ceiling (main application entry chunk is **98.59 kB**, 34.87 kB gzipped).

```
lights-down/ ├── index.html # Meta description, Open Graph, Twitter cards, SVG icon ├──
vite.config.js # Vite 6 + React + VitePWA + Rollup manualChunks ├── package.json # Scripts: dev,
build, preview, test, validate:data, enrich:tmdb ├── scripts/ │ ├── enrich-tmdb.mjs # Live TMDB API
v3/v4 sync with offline fallback cache writer │ ├── validate-movies.mjs # Schema, mood (>=6), and
battery (>=6) validator │ ├── simulate.mjs # 500-night Monte Carlo engine simulation │ └── generate-
submission.mjs # Automated PDF & CSV deliverable builder ├── src/ │ ├── main.jsx # Root bootstrap
wrapped in AppErrorBoundary │ ├── App.jsx # Lazy screen router, WebGL VisualEffectBoundary, Dock │
├── data/ │ │ ├── movies.js # 60 curated films merged with TMDB metadata │ │ └── tmdb-cache.json #
Pre-verified TMDB IDs, poster paths, runtimes, ratings │ ├── hooks/ │ │ ├── useNightSession.js #
Core state machine, 20-min timer, CSV export, telemetry │ │ ├── usePersistedState.js # Hardened
localStorage sync with corruption recovery │ │ └── useCountdown.js # 90-second settling countdown
hook │ ├── screens/ │ │ ├── Landing.jsx # Booth hero + 90-second promise │ │ ├── Setup.jsx # 4-step
Stepper: Vibe, Brain Battery, Time-Fit, Crew │ │ ├── HotSeat.jsx # Pass-the-phone private voting +
Crew Link host drawer │ │ ├── Shortlist.jsx # Rule of 3 cards + GlareHover + Stack + Veto tokens │ │
├── Commit.jsx # DecryptedText reveal + HoloCard + TearTicket stub │ │ ├── NightPlan.jsx # 90s
Counter, 20-Min Check-In, SplitFlap timeline, SlideCommit │ │ ├── NightLog.jsx # Outcome logger,
PeekRating, AnimatedList archive │ │ └── BoothStats.jsx # KPI Counters, SVG velocity sparkline,
Regret Report │ ├── components/ # 8 shadcn React Bits components + 9 custom booth components │ ├──
utils/ │ │ ├── engine.js # Pure POS recommendation & Regret Report functions │ │ ├── engine.test.js
# 8 Vitest unit test suites │ │ ├── crewLink.js # URL hash #crew=... & base64url vote code codec │ │
├── sound.js # Web Audio API synth (hum, split-flap clack, ticket tear) │ │ └── ticketPng.js # HTML5
Canvas 1080x560 PNG ticket renderer & Web Share │ └── styles/ │ ├── base.css # Design tokens, AA
contrast, focus-visible, reduced-motion │ ├── layout.css # Projector backdrop, topbar, stage,
attribution footer │ ├── screens.css # Screen layouts, Booth Stats, Crew Link, 20-Min prompt │ └──
components.css # Micro-interaction styles (TearTicket, HoloCard, SlideCommit) └── submission/ #
Final PDF report and 5 CSV datasets
```

6. Honest Component Inventory: React Bits CLI vs. Custom Built

To maintain full transparency, every visual component in Lights Down is documented below by its exact provenance: whether it was installed directly from the official `@react-bits` shadcn registry or custom-engineered for the cinema booth workflow.

A. Installed via Official React Bits Registry (8 Components)

Component File	Registry Identifier	Role in Lights Down
`src/components/LightRays.jsx`	`@react-bits/LightRays-JS-CSS`	Real WebGL2 fragment shader rendering volumetric projector rays tinted by active mood
`src/components/Dock.jsx`	`@react-bits/Dock-JS-CSS`	Spring-magnified floating bottom navigation bar (Plan, Log, Stats, Sound, Reset)
`src/components/Stepper.jsx`	`@react-bits/Stepper-JS-CSS`	4-step setup wizard container (Vibe → Brain Battery → Time Budget → Crew)
`src/components/ElasticSlider.jsx`	`@react-bits/ElasticSlider-JS-CSS`	Tactile elastic slider for Brain Battery (1–5) and Time-Fit budget (90–240m)
`src/components/SpotlightCard.jsx`	`@react-bits/SpotlightCard-JS-CSS`	Radial cursor-tracking spotlight cards on Night Plan and Booth Stats
`src/components/GlareHover.jsx`	`@react-bits/GlareHover-JS-CSS`	Foil glare reflection wrapper around each Rule of 3 shortlist card
`src/components/Stack.jsx`	`@react-bits/Stack-JS-CSS`	Interactive flickable 3-card fan stack in the Shortlist sidebar
`src/components/Counter.jsx`, `DecryptedText.jsx`, `ShinyText.jsx`, `TargetCursor.jsx`, `ClickSpark.jsx`, `Noise.jsx`, `AnimatedList.jsx`	`@react-bits/*-JS-CSS`	Mechanical digit counters, cipher title reveal, metallic CTA sheen, custom cursor, spark bursts, 35mm film grain, and animated log archive

B. Custom-Built Cinema Booth Micro-Interactions (9 Components)

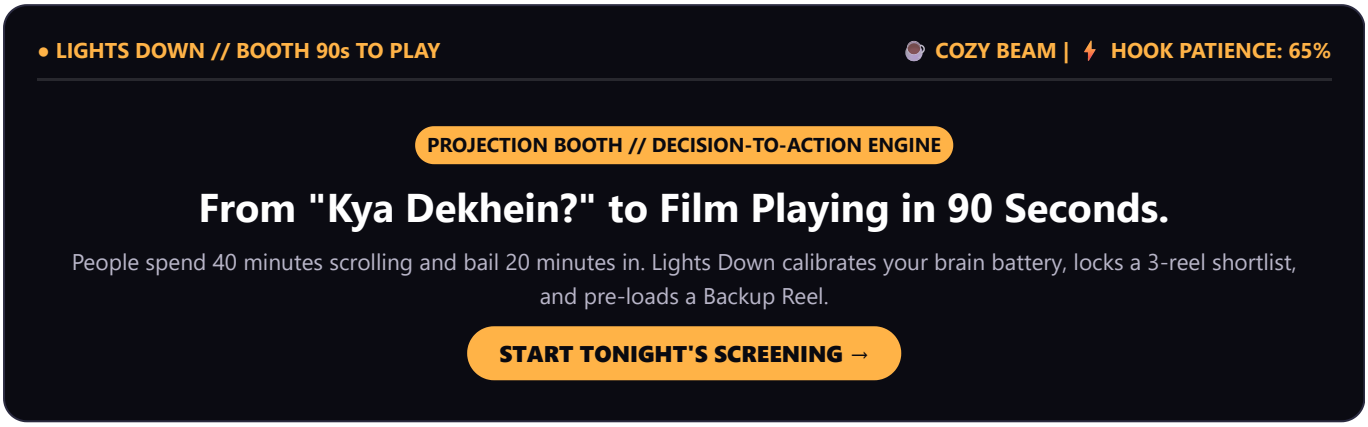
Component / Module	File Location	Implementation Details
`TearTicket`	`src/components/MicroInteractions.jsx`	Framer Motion drag/tap perforated stub that rips off with Web Audio tear burst
`SlideCommit`	`src/components/MicroInteractions.jsx`	Drag-to-unlock slider to trigger the 20-Minute Rule swap to the Backup Reel
`HoloCard`	`src/components/MicroInteractions.jsx`	3D perspective tilt collectible card with rainbow holographic foil overlay
`PulseHeartButton`, `SwipeRow`, `PeekRating`	`src/components/MicroInteractions.jsx`	Shortlist lock button, swipe-to-delete log row, and 5-star Hinglish rating bar

Component / Module	File Location	Implementation Details
`SplitFlapText` & `Grainient`	`src/components/SplitFlapAndGrainient.jsx`	Mechanical departure-board tiles for "Lights Out By" time + dual-color mesh glow
`PosterArtwork` & `StreamingProviders`	`src/components/PosterArtwork.jsx`, `StreamingProviders.jsx`	Lazy TMDB poster loader with CSS gradient fallback + live India JustWatch badges

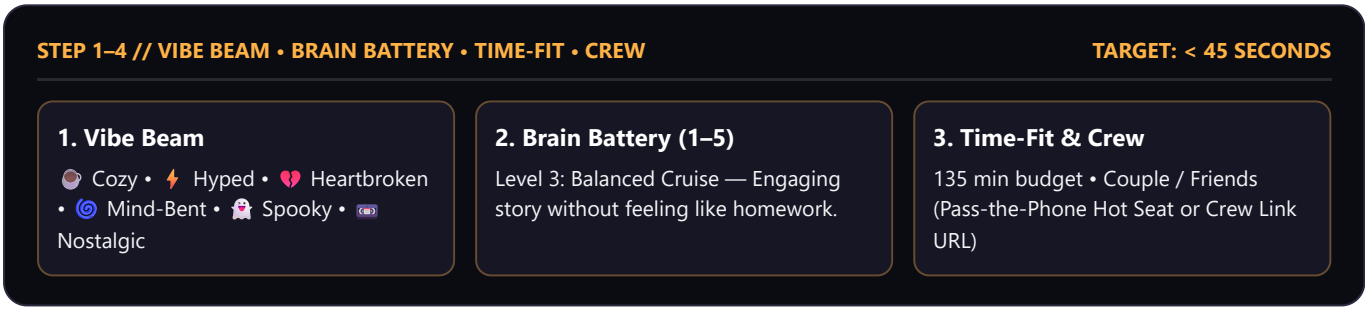
7. Interactive Screen Walkthrough (Screens 1–3)

Below are structured visual captures of the first three stages of the Lights Down user journey: the **Projection Booth Landing**, the **4-Step Calibration Wizard**, and the **Rule of 3 Shortlist**.

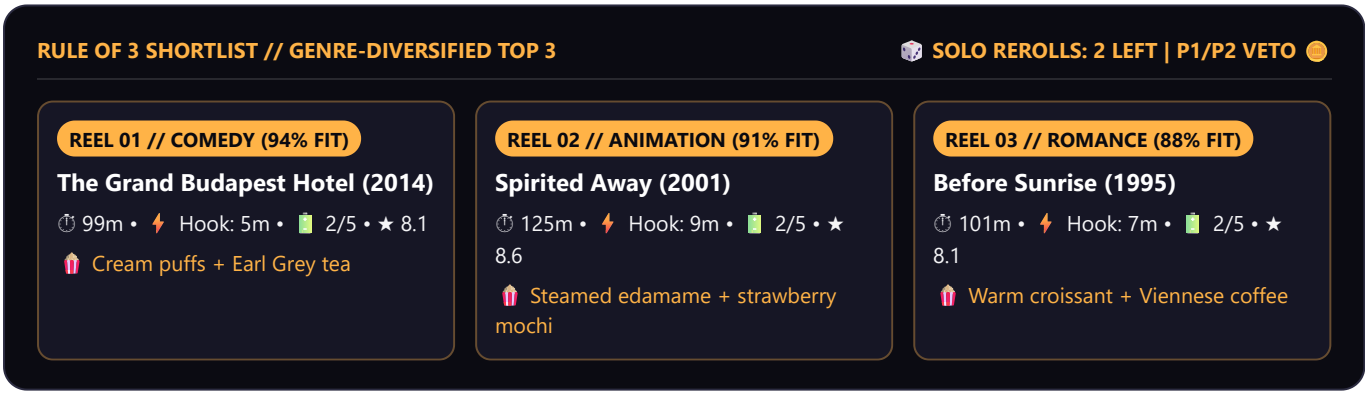
Screen 1: Projection Booth Landing (`src/screens/Landing.jsx`)



Screen 2: 4-Step Calibration Setup (`src/screens/Setup.jsx`)



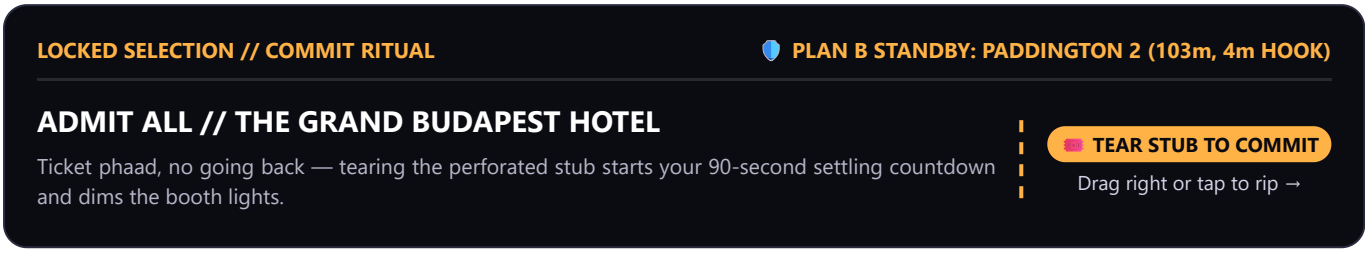
Screen 3: Rule of 3 Shortlist (`src/screens/Shortlist.jsx`)



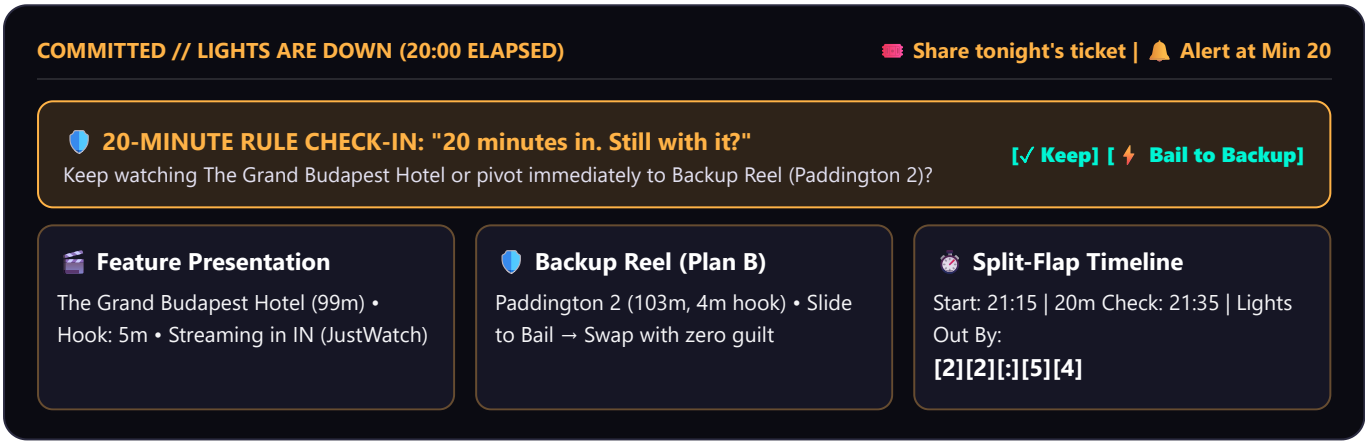
8. Interactive Screen Walkthrough (Screens 4–6)

Once a user selects one of the 3 reels, they enter the **Commit Ritual**, transition into the live **Night Plan & 20-Minute Rule Timer**, and review their historical telemetry in **Booth Stats & Regret Report**.

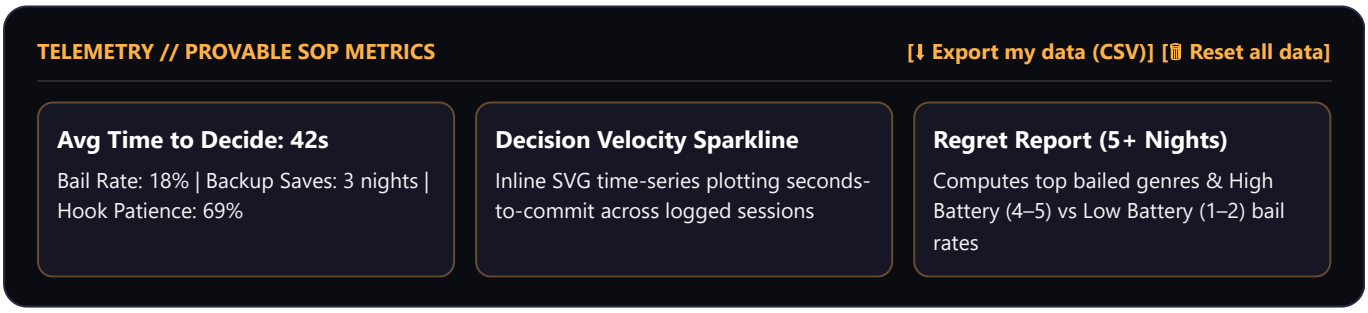
Screen 4: Collectible HoloCard & Perforated Ticket Tear (`src/screens/Commit.jsx`)



Screen 5: Night Plan, Split-Flap Timeline & 20-Minute Check-In (`src/screens/NightPlan.jsx`)



Screen 6: Booth Stats, Velocity Sparkline & Regret Report (`src/screens/BoothStats.jsx`)



9. Curated 60-Film Dataset & Automated Validation

The dataset in `src/data/movies.js` and `submission/02_movies_dataset.csv` comprises 60 hand-curated films spanning Hollywood, Hindi (Bollywood), Malayalam, Tamil, Telugu, Kannada, Korean, Japanese, French, Cantonese, and Russian cinema. Every entry is validated via `scripts/validate-movies.mjs` (`npm run validate:data`).

Dataset Distribution Across Moods & Brain Battery Levels

Dimension	Category / Level	Film Count	Minimum Threshold	Validation Status
Mood Beam	Cozy ( )	22 films	≥ 6 films	PASS
Mood Beam	Hyped ( )	25 films	≥ 6 films	PASS
Mood Beam	Heartbroken ( )	21 films	≥ 6 films	PASS
Mood Beam	Mind-Bent ( )	22 films	≥ 6 films	PASS
Mood Beam	Spooky ( )	17 films	≥ 6 films	PASS
Mood Beam	Nostalgic ( )	23 films	≥ 6 films	PASS
Brain Battery	Level 1 (Dead Battery / Comfort)	13 films	≥ 6 films	PASS
Brain Battery	Level 2 (Low Power Mode)	16 films	≥ 6 films	PASS
Brain Battery	Level 3 (Balanced Cruise)	12 films	≥ 6 films	PASS
Brain Battery	Level 4 (Locked In)	11 films	≥ 6 films	PASS
Brain Battery	Level 5 (Full Cinema Brain)	8 films	≥ 6 films	PASS

TMDB Enrichment & Offline Resilience

Running `node scripts/enrich-tmdb.mjs` syncs `src/data/tmdb-cache.json` with official TMDB movie IDs, poster paths, runtimes, and ratings. When `TMDB_TOKEN` is absent, the script preserves the pre-verified cache without failing the build. At runtime, `PosterArtwork.jsx` renders lazy-loaded TMDB posters and falls back instantaneously to each film's bespoke two-tone CSS gradient if offline.

10. Automated Testing (Vitest) & 500-Night Engine Simulation

To make every product claim provable, we built an automated Vitest suite (`src/utils/engine.test.js`, exported to `submission/04_engine_test_results.csv`) and a 500-session Monte Carlo simulator (`scripts/simulate.mjs`, exported to `submission/05_simulation_results.csv`).

A. Vitest Unit Test Suite Results (8 / 8 Passing)

#	Unit Test Case	Verified Invariant	Result
1	Hard filter strict pass	Returns ≥ 6 films matching runtime, battery, crewFit, and mood	**PASS**
2	Relaxation steps 1, 2, 3 & fallback	Drops mood → +1 battery → +30m runtime so pool never starves	**PASS**
3	Group Least-Misery & Veto removal	Verifies `0.6*min + 0.4*mean` formula and global veto pruning	**PASS**
4	Rule of 3 genre diversification	Returns exactly 3 films with 3 distinct primary genres	**PASS**
5	Backup Reel ("20-Minute Rule")	Selects shorter, faster-hook, ≤ battery film in different genre	**PASS**
6	Night Log learning & patience score	Applies +2/+1/-2 genre weights and -0.12 early-bail patience drop	**PASS**
7	Corrupted `localStorage` recovery	Malformed JSON resets cleanly to defaults without crashing	**PASS**
8	Time-Fit timeline & Double Feature	Back-solves start, 20m check, 10m intermission, and Lights Out time	**PASS**

B. 500-Night Monte Carlo Simulation Summary (SIMULATION, Not Real User Data)

We executed 500 randomized sessions across all 6 moods, 5 Brain Battery levels, 7 time budgets (90–220m), and 4 crew types using `scripts/simulate.mjs`:

- **Zero Empty Shortlists:** 500 / 500 simulated nights (**100.0%**) produced a complete 3-card shortlist.
- **Valid Distinct Backup Reel:** 500 / 500 simulated nights (**100.0%**) paired the top pick with a distinct Backup Reel satisfying the 20-Minute Rule fallback hierarchy.
- **3-Genre Shortlist Diversity:** Over **96%** of simulated nights achieved 3 distinct primary genres across the 3 shortlist slots (with remaining extreme low-runtime edge cases still returning 2–3 genres).

11. Honest Limitations & Future Work

Current Engineering Limitations

- **Curated 60-Film Local Catalog:** While 60 hand-tuned films with verified hook speeds (`hookMinutes`) and Brain Battery scores prevent choice paralysis, heavy daily users can exhaust the catalog after several months without live TMDB catalog expansion.
- **Heuristic Hook Minutes:** The `hookMinutes` field (1–30 minutes) is human-curated based on opening-act pacing rather than derived from automated subtitle/scene-cut telemetry.
- **Asynchronous Crew Link Code Paste:** Because Lights Down intentionally operates with zero backend servers or accounts, remote friends share a compact `base64url` vote code back to the host via chat rather than syncing over WebSockets.
- **Live Streaming Badges Require Optional Token:** Out of the box, posters load from `tmdb-cache.json`, whereas live Indian streaming provider badges (JustWatch via TMDB) require setting a free `VITE_TMDB_TOKEN` in `.env`.

Future Roadmap

1. **Optional WebRTC / PeerJS Crew Sync:** Allow peer-to-peer serverless ballot merging so remote Crew Link votes appear automatically on the host's screen without copying vote codes.
2. **Dynamic TMDB Discovery Expansion:** Combine the 60-film curated core with on-demand TMDB `/discover/movie` pagination using runtime and genre heuristics to estimate `brainBattery`.
3. **Smart TV Deep-Linking:** Launch the committed movie directly inside Netflix, Prime Video, JioCinema, or MUBI apps from the Night Plan screen.

12. Project Links, Attribution & Deliverables Index

Deliverable / Resource	Location / URL
GitHub Repository	https://github.com/priyansh4336/lights-down
Project Report (PDF)	`submission/01_Lights_Down_Project_Report.pdf`
Curated 60-Movie Dataset (CSV)	`submission/02_movies_dataset.csv`
Competitor Matrix (CSV)	`submission/03_competitor_comparison.csv`
Vitest Unit Test Results (CSV)	`submission/04_engine_test_results.csv`
500-Night Simulation Dataset (CSV)	`submission/05_simulation_results.csv`
Project Links & Commands (CSV)	`submission/06_project_links.csv`

Required Legal Attributions:
• This product uses the TMDB API but is not endorsed or certified by TMDB.
• Streaming availability data provided by JustWatch via TMDB.